

Response to Reviewer Comments

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A New IoT Platform Based on PQC Algorithms: A Solution for Quantum Computing Threats

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We thank the reviewer for the careful assessment and for recognizing the paper’s practical treatment of post-quantum cryptography (PQC) migration in heterogeneous Internet of Things (IoT) systems. We agree that the submitted figures were not suitable for publication. We replaced the four submitted figures with two author-redrawn vector schematics. Their SVG sources are included with the manuscript, and LaTeX uses cropped vector-PDF versions. The title and author list are unchanged, as are all reported benchmark values and the central placement conclusion. The revision also narrows the deployment claims to the evidence from the QEMU experiments, clarifies node and gateway responsibilities, removes the draft’s unevaluated DMA and design-for-test details, and updates the supporting references.

Response to the overall assessment

Reviewer assessment: *Borderline Accept. The paper’s core strength lies in its practical, architecture-driven approach to migrating heterogeneous IoT ecosystems to PQC. The study introduces a strategic algorithm-placement model across device tiers.*

Response. Section III-A now states the node, gateway, and server responsibilities as an explicit tiered placement model. Table I gives the placement and rationale for ML-KEM, ML-DSA, SLH-DSA, and hybrid mode. The Introduction uses the same framing, and the Conclusion limits the deployment claim to the node/gateway workflow measured in the prototype.

Response to the figure-quality comment

Reviewer comment: *I am mainly concerned about the AI-generated figures (Fig. 1 and Fig. 4). The human-generated figures (Fig. 2 and Fig. 3) are of low quality and therefore must be improved.*

Response. We agree. None of the four submitted image files is used in the revised manuscript. We redrew and consolidated the overview, hardware, sequence, and deployment views into two vector schematics. Both figures consist of vector paths and contain no embedded raster images. They use consistent line weights, typography, arrow directions, component boundaries, and captions.

The two figures divide the content as follows:

- **Fig. 1, “End-node architecture for PQC-enabled IoT devices,”** shows the processing core, PQC software stack, bootloader and secure boot, RAM, storage, network path, system bus, and peripheral interfaces discussed in Section III-C.
- **Fig. 2, “Three-tier deployment map for quantum-safe session establishment and passkey authentication,”** shows constrained and Linux-capable end nodes, gateway-side ML-KEM/KDF/AEAD and signature functions, optional direct PQC operation, and cloud/server trust services discussed in Section III-D.

The sequence previously shown in submitted Fig. 3 is now described in Section III-D. The revised Fig. 2 combines the tier-placement content that was previously split between the overview and deployment figures. The manuscript contains exactly two figures and has no obsolete references to Figs. 3 or 4. The accompanying text clarifies placement conditions, credential ownership, gateway delegation, and failure handling. The reported benchmark values are unchanged, while the deployment claims have been narrowed to what the QEMU prototype supports.

Response to technical quality and soundness

Reviewer assessment: *Fair (Some technical issues or missing baselines, but potentially fixable).*

Response. We did not have a matched physical-board or accelerator baseline for this revision and therefore did not add an unmeasured comparison. Section IV now identifies the reported values as software measurements from a QEMU prototype, used as evidence for placement decisions rather than as cycle-accurate hardware results. Section IV-C states that Linux/QEMU resident-set size is not an estimate of ESP32 SRAM use. The limitations paragraph assigns physical-board power, radio throughput, flash wear, timing leakage, and comparison with existing PQC IoT architectures to future work.

Table I connects each primitive to a deployment tier and explains the placement. Table II reports elapsed time, resident memory,

payload size, and the recommended tier; its measured values are unchanged. Section IV-D relates those measurements to buffer sizing, gateway admission control, and possible acceleration kernels. The end-node description identifies AXI, AHB, and APB as interfaces of the memory-mapped system bus. It replaces the draft’s DMA-specific data path with a platform-neutral shared-buffer model because the evaluation did not measure a DMA implementation. The secure-channel text names AES-GCM and ChaCha20-Poly1305 as example authenticated-encryption schemes. We also removed the draft’s design-for-test and fault-injection extension because it was outside the implemented and measured system.

Response to originality and novelty

Reviewer assessment: *Fair (Incremental improvement or a minor new application).*

Response. The paper does not claim a new cryptographic primitive. The revised Introduction defines the contribution as a layered node/gateway/server architecture, a PQC-aware passkey metadata and hybrid-policy model, and a measurement-based rule for placing ML-KEM, ML-DSA, and SLH-DSA. Section III and the two revised figures follow these system-level contributions. The standardized algorithms and reference software remain the basis of the prototype.

Summary of manuscript changes

Reviewer concern	Revision	Location
AI-generated appearance in submitted Figs. 1 and 4	Replaced the submitted artwork with two author-redrawn SVG schematics and vector-PDF derivatives	Revised Figs. 1 and 2; Sections III-C–D
Low quality of submitted Figs. 2 and 3	Standardized line art, typography, arrows, labels, and captions; removed redundant figure content	Revised Figs. 1 and 2
Technical scope and possible missing baselines	Kept all measured values; narrowed the deployment claim to the QEMU evidence; replaced the unevaluated DMA-specific path with the shared-buffer model; removed the design-for-test extension; stated future work	Sections III-C–E; IV-A–D; Conclusion
Security boundaries and placement policy	Clarified private-key ownership, gateway delegation, hybrid-policy enforcement, and the conditions for node or gateway placement	Sections III-A–E
Migration evidence and references	Corrected bibliographic data, removed the ML-DSA-for-WebAuthn Internet-Draft citation, and added transition and fragmentation sources used in the revised discussion	Sections II-B–D; References
Incremental novelty	Stated the contribution as tiered placement, migration metadata/policy, and measurement-guided system design rather than a new primitive	Introduction; Section III-A; Conclusion

We thank the reviewer again. The camera-ready manuscript contains the final vector artwork and the corresponding figure references. We added no experiment or numerical result, and all reported benchmark values are unchanged. The other revisions clarify the placement policy, security boundaries, limitations, and supporting references.